

Adelic Shannon Theory: A Research Design

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Abstract

Standard Shannon theory — channel capacity, entropy, source coding, and the Gaussian noise model — is archimedean: it measures information in bits over $\mathbb{R}$. The Adelic Physics Programme’s central claim — $\mathbb{Q}$, not $\mathbb{R}$, is the physically accessible base field — suggests that information theory itself may have a p-adic component invisible to the archimedean formulation. We propose the foundational sketch of an Adelic Shannon Theory: a generalisation of information theory to the adèle ring $\mathbb{A}_{\mathbb{Q}}$. Three components are specified: (1) p-adic entropy — replacing the standard logarithm with the p-adic valuation as the uncertainty measure; (2) the adelic channel capacity — a product over all places $\prod_p C_p \times C_{\infty}$, with a conjectured product-formula coding theorem; and (3) the Poisson summation formula as the adelic source coding theorem — the Gaussian $\hat{e}^{-\pi x^2}$, uniquely invariant under the adelic Fourier transform, is the maximum-entropy distribution at every place. The research design includes methodology sketches, effort estimates (18 months), connections to existing QNFO infrastructure (Ultrametric Engine, p-adic valuation classifiers), and falsifiability conditions. This paper does not construct the theory; it poses the problem and specifies the research programme. [SPECULATIVE]

Keywords: adelic Shannon theory, p-adic entropy, channel capacity, Poisson summation, Gaussian, adèle ring, information theory, ultrametric noise, source coding

1. Introduction: The Archimedean Assumption in Information Theory

1.1 What Shannon Assumed

Claude Shannon’s 1948 “A Mathematical Theory of Communication” [1, established] founded information theory on two archimedean pillars:

1. **Entropy:** $H(X) = -\sum p(x) \log_2 p(x)$, where the logarithm is the standard real-valued logarithm over $\mathbb{R}$.
2. **Channel capacity:** $C = \max_{\{p(x)\}} I(X; Y)$ for a channel defined by a transition probability matrix $p(y|x)$ over a finite (or continuous) alphabet — always measured in bits, a real number.

In the continuous case, the additive white Gaussian noise (AWGN) channel has capacity $C = \frac{1}{2} \log_2(1 + \text{SNR})$, and the Gaussian distribution is the maximum-entropy distribution given a fixed variance.

Shannon’s formulation makes no explicit commitment to $\mathbb{R}$ as the base field — but the continuous channel capacity formula, the differential entropy definition, and the Gaussian noise model all assume an archimedean framework: probabilities are real numbers, signals are real-valued functions of time, and noise is additive with a real-valued Gaussian distribution.

1.2 The $\mathbb{Q}$ Challenge

The Adelic Physics Programme [2] argues that $\mathbb{Q}$, not $\mathbb{R}$, is the physically accessible base field. If true, this challenges the archimedean assumption in information theory:

- **Signals are rational:** Any physical signal, when measured, yields a finite-precision rational number — a tally of detector clicks, a voltage quantised by the ADC resolution. The signal is not a real-valued function; it is a rational-valued sequence.
- **Noise is not Gaussian:** The AWGN model is a continuous idealisation. Physical noise, when measured at sufficient precision, reveals discrete structure — Poisson shot noise, quantisation error, p-adic valuations of the noise covariance matrix.
- **Channel capacity is a real number:** $C = \frac{1}{2} \log_2(1 + \text{SNR})$ is a theorem about real-valued signals. If signals are fundamentally rational, the capacity may have a p-adic component.

1.3 This Paper

This paper does NOT construct Adelic Shannon Theory. It poses the problem. Specifically:

1. **p-adic entropy:** Define $H_p(X) = -\sum p(x) v_p(x)$, where v_p is the p-adic valuation. What are the properties? Does this satisfy the Shannon axioms?
2. **Adelic channel capacity:** Conjecture that the capacity $C(\mathbb{A}_p) = \prod_p C_p \times C_\infty$. A product-formula coding theorem would state that reliable communication is possible at any rate below the product capacity and impossible above.
3. **Poisson summation as source coding:** Show that the Gaussian $\hat{e}^{-\pi x^2}$ — the unique function invariant under the adelic Fourier transform — is the maximum-entropy source distribution at every place simultaneously.

The paper provides methodology sketches, connects to existing QNFO infrastructure, and specifies falsifiability conditions.

2. p-Adic Entropy

2.1 The Standard Definition

Shannon entropy [1, established] for a discrete random variable X with probability mass function $p(x)$ is:

$$H(X) = -\sum_{x \in \Omega} p(x) \log_2 p(x)$$

The logarithm is over $\mathbb{R}$ — specifically, $\log_2$ is the natural logarithm scaled by $1/\ln(2)$. The entropy is measured in bits, a dimensionless real number.

2.2 The p-Adic Generalisation

[SPECULATIVE] For a random variable X taking values in $\mathbb{Q}_p$ (or more practically, in $\mathbb{Z}$, interpreted p-adically), define the **p-adic entropy**:

$$H_p(X) = -\sum_{x \in \Omega} p(x) v_p(x)$$

where $v_p(x)$ is the p-adic valuation — the exponent of the highest power of p dividing x .

Properties (conjectured):

1. **Non-negativity:** $H_p(X) \geq 0$ (true since $v_p(x) \geq 0$ for integer x , and $p(x) \leq 1$).
2. **Additivity:** $H_p(X, Y) = H_p(X) + H_p(Y)$ for independent X, Y ? Not generally true — $v_p(x + y)$ is not additive. The p-adic valuation satisfies $v_p(x \cdot y) = v_p(x) + v_p(y)$ and the strong triangle inequality $v_p(x + y) \geq \min(v_p(x), v_p(y))$. The additivity property of entropy may need to be modified for p-adic entropies.
3. **Maximum entropy:** Which distribution maximises $H_p(X)$ subject to a constraint on the expected valuation $E[v_p(X)]$? The answer is not known.

2.3 Interpretation

The p-adic entropy measures the “uncertainty in the p-adic structure” of a random variable, not its uncertainty in the Shannon sense. A uniform distribution over $\{0, \dots, 2^k - 1\}$ has maximal Shannon entropy (k bits) but may have low p-adic entropy if most numbers have small p-adic valuation.

This is a structural subtlety: in the adelic framework, information is a vector $(H_\infty, H_2, H_3, H_5, \dots)$ — the Shannon entropy at the archimedean place, and the p-adic entropies at the non-archimedean places. The “total information” is not a single number but a product over all places.

2.4 Effort Estimate

Subproblem	Effort	Difficulty
Formalise p-adic entropy axioms	2 months	Moderate
Prove properties (additivity, maximum entropy, data-processing inequality)	6 months	High (unknown territory)
Connect to existing p-adic probability theory (Khrennikov, Vladimirov, Volovich)	2 months	Moderate
Total	~10 months	—

3. Adelic Channel Capacity

3.1 The Standard Capacity

For the AWGN channel with signal-to-noise ratio SNR, the capacity is:

$$C = \frac{1}{2} \log_2(1 + \text{SNR}) \text{ bits}$$

This is a theorem: for any $R < C$, there exists a code achieving error probability $\rightarrow 0$; for any $R > C$, no such code exists. [established — Shannon 1948]

3.2 The Adelic Conjecture

[SPECULATIVE] For an adelic channel whose input and output are adeles (i.e., vectors $(x_\infty, x_2, x_3, \dots)$ with an RN component and Q_p components for each prime p), the capacity factorises:

$$C(A_Q) = C_\infty \times \prod_p C_p$$

where: - $C_\infty = \frac{1}{2} \log_2(1 + \text{SNR}_\infty)$ is the standard archimedean capacity (bits per channel use)
- C_p is the p -adic capacity for noise at the p -adic place (units: p -adic bits per channel use)

The **product-formula coding theorem**: there exists an adelic code achieving reliable communication at any rate below the product capacity, and no code exists above.

3.3 The Ultrametric Noise Model

The p -adic analogue of the AWGN channel is the **Additive Ultrametric (AUM) channel**: the noise is a p -adic random variable with a distribution peaked at 0 (small valuation) and decaying as v_p increases:

$$p(n) \propto p^{-\{v_p(n)\}} \text{ for } n \in \mathbb{Z}_p$$

The AUM capacity is conjectured to be:

$$C_p = \log_p(1 + \text{SNR}_p) \text{ “p-adic bits”}$$

where SNR_p is the ratio of the signal valuation scale to the noise valuation scale. This has NOT been proved — it is a conjecture motivated by formal analogy with the AWGN case.

3.4 Effort Estimate

Subproblem	Effort	Difficulty
Define AUM channel rigorously	2 months	Moderate
Prove $C_p = \log_p(1 + \text{SNR}_p)$	6 months	High (unproved conjecture)
Product-formula coding theorem	6 months	Highest
Total	~14 months	—

4. Poisson Summation as Source Coding

4.1 The Archimedean Maximum-Entropy Principle

In standard information theory [1, established], the Gaussian distribution maximises differential entropy subject to a fixed variance constraint:

$$\max_{\{p(x): E[X^2] = \sigma^2\}} h(X) = \frac{1}{2} \log_2(2\pi e \sigma^2)$$

achieved by $X \sim N(0, \sigma^2)$. [established — Cover & Thomas 2006]

This is the **source coding interpretation**: among all sources with fixed average power, the Gaussian source requires the highest channel capacity to transmit — it is the “most random” source.

4.2 The Adelic Source Coding Conjecture

[SPECULATIVE] The Gaussian $e^{-\pi x^2}$ is not merely the maximum-entropy distribution at the archimedean place. It is the maximum-entropy distribution **at every place simultaneously** — the unique function whose entropy is maximal both in the standard Shannon sense (archimedean) and in the p-adic valuation sense (non-archimedean, with the characteristic function of $\mathbb{Z}_p$ playing the p-adic role).

The Poisson summation formula:

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{n \in \mathbb{Z}} \hat{f}(n)$$

is then interpreted as a **source coding theorem** on $A_{\mathbb{Q}}$:

- The LHS (sum over $\mathbb{Z}$) is the source — a discrete lattice of signal values, each encoded as a rational number.
- The RHS (sum over dual lattice) is the channel — the Fourier-transformed signal received at the destination, with the same rational structure.
- The Gaussian is the optimal source distribution — the only distribution for which the source entropy and channel entropy are equal (invariance under the adelic Fourier transform).

The equality is exact — not an inequality, not an asymptotic bound. This is a stronger statement than the standard source-channel separation theorem in information theory: the Gaussian achieves equality of source and channel entropies, not merely a capacity bound.

4.3 Connection to Existing Infrastructure

The QNFO Ultrametric Engine [3] provides spectral analysis endpoints that compute Tate, Amice, and intrinsic Amice transforms — the adelic spectral toolkit. Extending these endpoints to compute adelic entropies and capacities would provide the computational backbone for testing the Poisson summation source coding conjecture.

4.4 Effort Estimate

Subproblem	Effort	Difficulty
Formalise the Poisson sum-to-source coding mapping	3 months	Moderate
Prove Gaussian is maximum-entropy at all places	6 months	High (partial: known for $\mathbb{R}$ and $\mathbb{Q}_p$ separately; global result not proved)
Connect to adelic Fourier analysis (Tate thesis)	3 months	Moderate
Total	~12 months	—

5. Overall Programme

5.1 Timeline

Component	Effort	Dependencies
p-adic entropy	10 months	None
Adelic channel capacity	14 months	p-adic entropy
Poisson sum source coding	12 months	Adelic channel capacity
Total	~18 months (parallelised)	—

5.2 Success Criteria

1. **p-adic entropy:** $H_p(X)$ satisfies the Shannon axioms (non-negativity, additivity for independent variables, subadditivity, concavity with respect to the probability distribution) with appropriate generalisations.
2. **Product-formula coding theorem:** A reliable adelic code exists for any rate below $C(A_\mathbb{Q})$ and no code exists above.
3. **Poisson sum source coding:** The Gaussian $e^{\{-\pi x^2\}}$ is rigorously proved to be the maximum-entropy source at all places simultaneously.

5.3 Falsifiability

[UNTESTED] The adelic Shannon programme is falsified if:

1. The p-adic entropy $H_p(X)$ does NOT satisfy a generalised data-processing inequality — i.e., there exists a transformation T such that $H_p(T(X)) > H_p(X)$, meaning “information can be created by processing” — which would violate a fundamental axiom of information theory.
2. The product-formula coding theorem does NOT hold — i.e., there exists a rate $R < C(A_\mathbb{Q})$ for which no reliable code exists, or (less likely) a rate $R > C(A_\mathbb{Q})$ for which a reliable code does exist.
3. The Gaussian is NOT the maximum-entropy source at a specific p-adic place — i.e., there exists a distribution on $\mathbb{Q}_p$ with the same valuation variance as the characteristic function of $\mathbb{Z}_p$ with strictly higher p-adic entropy.

6. Connection to the Adelic Programme

Adelic Shannon Theory connects to every strand of the wider Adelic Physics Programme [2]:

Paper	Connection
P3 (Poisson Summation) [4]	The mathematical foundation — Poisson summation IS the source-channel equality in the adelic framework
P5 (FACTORING + Adelic Complexity) [5]	Silent-radix cryptography as an instance of place-dependent information content — the Shannon capacity of SRE is a product over places
P6 (Falsifiability Protocol) [6]	The ultrametric clustering signature (S2) can be reinterpreted as an information-theoretic measurement: is the channel between the quantum system and the classical measurement apparatus p-adic (AUM) or archimedean (AWGN)?

Paper	Connection
P8 (p-Adic QEC Verification) [7]	The Mahler v_p spectrum is, in the adelic Shannon framework, the p-adic information content of the code — the QEC classifier is an adelic Shannon entropy classifier

7. Conclusion: Beyond Bits

Shannon taught us to measure information in bits. A bit is an archimedean unit — one binary decision, a choice between 0 and 1 in $\mathbb{R}$. The adelic framework suggests that information may be measured in **p-adic bits** — choices in the p-adic valuation — and that the total information is a product over all completions of $\mathbb{Q}$.

The Poisson summation formula is the mathematical bridge that connects these measurements at different places. The Gaussian is the unique distribution that maximises information at every place simultaneously. The product-formula coding theorem, if proved, would state that reliable communication is possible at the product rate and impossible above — a generalisation of Shannon’s channel coding theorem to the adeles.

None of this has been proved. This paper is the problem statement — the invitation to construct the theory.

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Code Availability: The theory described here, if constructed, would integrate with the QNFO Ultrametric Engine [3].

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